Report

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Notes

Contents

1 Back to the Drawing Board - Removing Thrust as a Control **2**
1.1 Put Cross Product Evaluation as Discrete Constraint . . . 2
1.1.1 Setup . . . 2
1.1.2 Results . . . 2
1.1.3 Plots . . . 3
1.2 Remove the Integral . . . 5
1.2.1 Setup . . . 5
1.2.2 Results . . . 5
1.2.3 Plots . . . 6
1.3 No Optimization . . . 8
1.3.1 Setup . . . 8
1.3.2 Results . . . 8
1.3.3 Plots . . . 9
1.4 Minimize Final Time . . . 11
1.4.1 Setup . . . 11
1.4.2 Results . . . 11
1.4.3 Plots . . . 12
1.5 Cross Product as Objective . . . 14
1.5.1 Setup . . . 14
1.5.2 Results . . . 14
1.5.3 Plots . . . 15
1.6 Cross Product Only . . . 17
1.6.1 Setup . . . 17
1.6.2 Results . . . 17
1.6.3 Plots . . . 18
1.7 Notes . . . 20

2 Generating Tilted Orbits for (1,1) Cyclers **20**
2.1 45 Degree Guess . . . 20
2.1.1 Setup . . . 20
2.1.2 Results . . . 20
2.1.3 Plots . . . 21
2.2 60 degree Guess . . . 22
2.2.1 Setup . . . 22
2.2.2 Results . . . 22
2.2.3 Plots . . . 23
2.3 30 Degree Guess . . . 24
2.3.1 Setup . . . 24
2.3.2 Results . . . 24
2.3.3 Plots . . . 25
2.4 Notes . . . 26

1 Back to the Drawing Board - Removing Thrust as a Control

1.1 Put Cross Product Evaluation as Discrete Constraint

1.1.1 Setup

- Objective: $W_{cf} = W_c = 1$
  - $J = W_{cf}\left(C_f - C^*\right)^2 + \int_0^{t_f} W_c\left(C - C^*\right)^2 dt$
- Constraints: Let $\begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix} = \vec{v}_i \times \vec{v}_f$; $\epsilon = 1 \times 10^{-3}$
  - $x_f - x_i = 0$
  - $y_f - y_i = 0$
  - $z_f - z_i = 0$
  - $|c_1| \leq \epsilon$
  - $|c_2| \leq \epsilon$
  - $|c_3| \leq \epsilon$
- Bounds:
  - $t_0 = 0$
- Mesh:
  - Segments: 100
  - Points: 10
- Ipopt Options:
  - ipopt_options.tol: 1×10^{-8}

1.1.2 Results

- Period Error:
  - 158.05 [s]
  - 0.004087 [%]
- Jacobi Constant/Hamiltonian Deviation:
  - Maximum: 1.8987×10^{-10}
  - From C^*: 1.3831×10^{-8}
- Hamiltonian Deviation:
  - Maximum: 5.8505×10^{-9}
- Solution Run Time:
  - 28.0 [s]

1.1.3 Plots

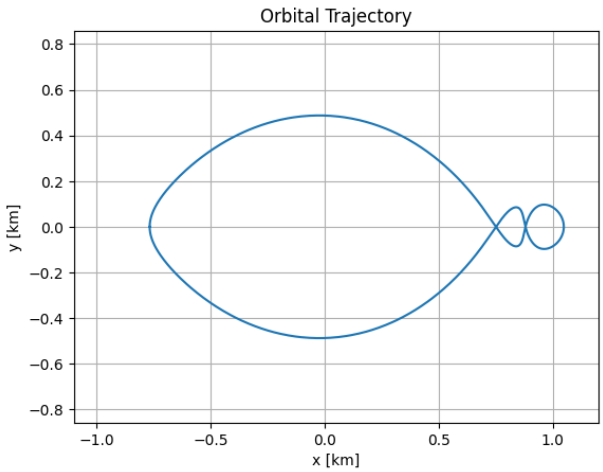

Figure 1: 2D Representation of Orbit (xy-plane)

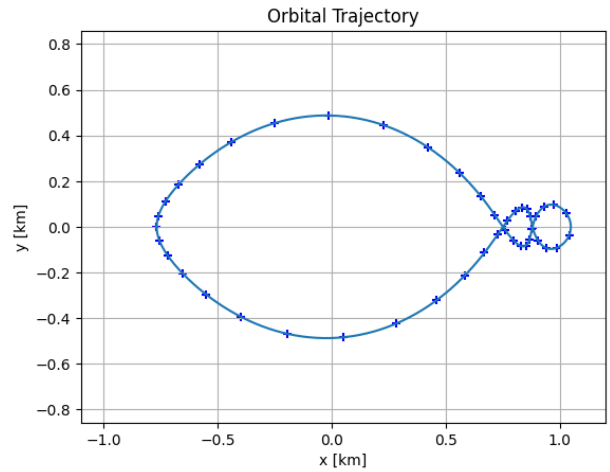

Figure 2: 2D Representation of Orbit with Day Ticks(xy-plane)

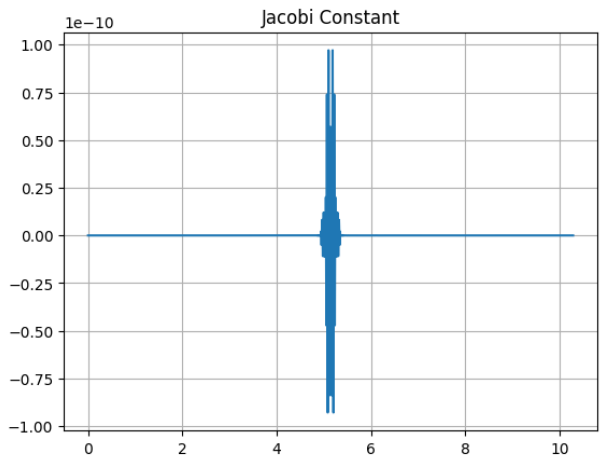

Figure 3: Jacobi Constant over Orbit

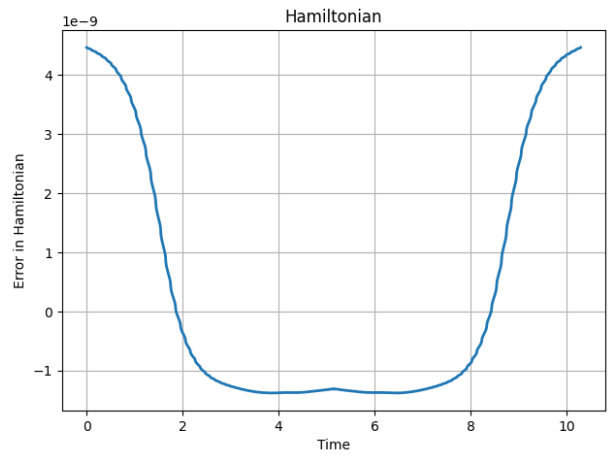

Figure 4: Hamiltonian over Orbit

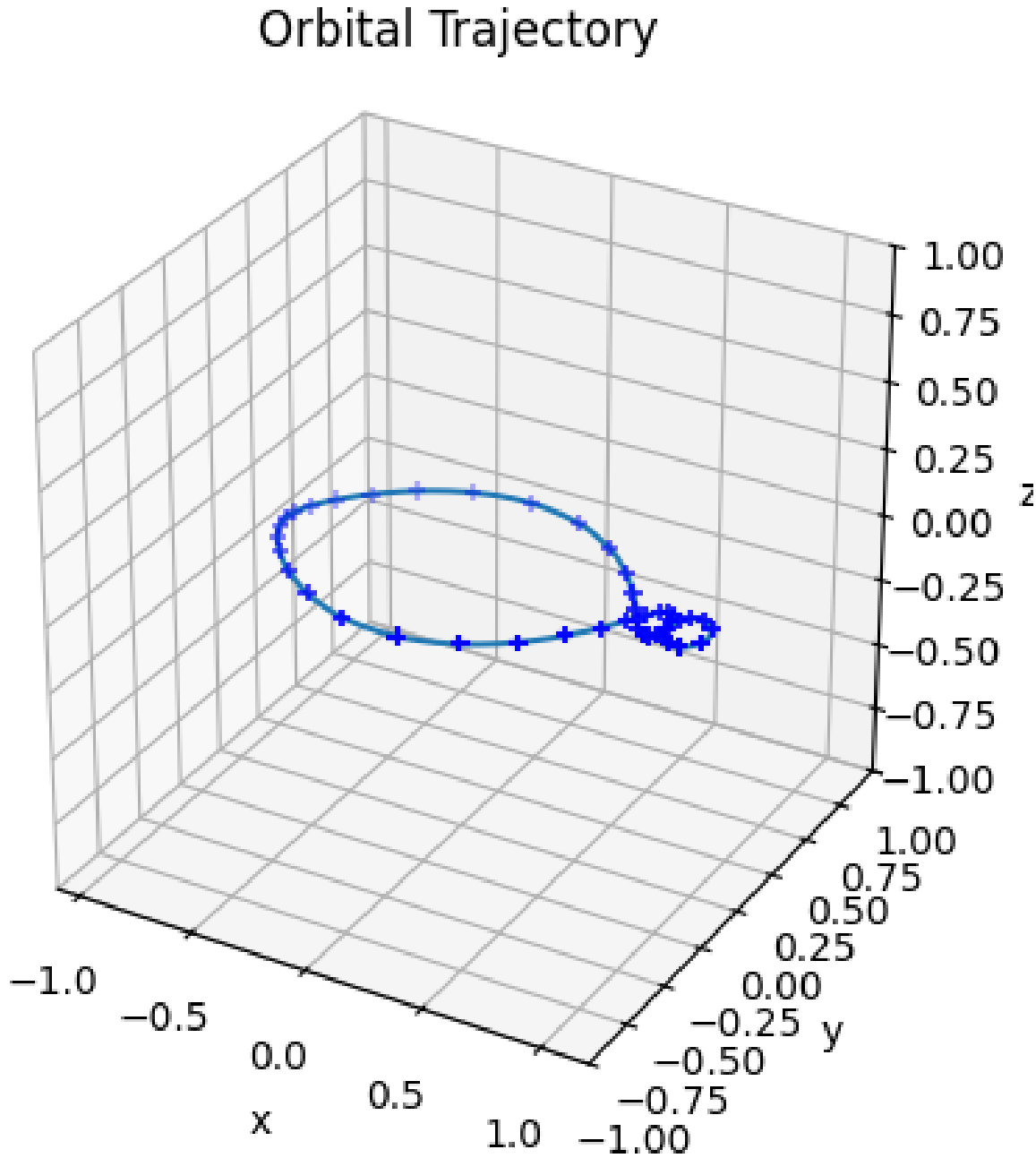

Figure 5: 3D Representation of Orbit

1.2 Remove the Integral

1.2.1 Setup

- Objective: $W_{cf} = 1$
  - $J = W_{cf} \left(C_f - C^*\right)^2$
- Constraints: Let $\begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix} = \vec{v}_i \times \vec{v}_f$; $\epsilon = 1 \times 10^{-3}$
  - $x_f - x_i = 0$
  - $y_f - y_i = 0$
  - $z_f - z_i = 0$
  - $|c_1| \leq \epsilon$
  - $|c_2| \leq \epsilon$
  - $|c_3| \leq \epsilon$
- Bounds:
  - $t_0 = 0$
- Mesh:
  - Segments: 100
  - Points: 10
- Ipopt Options:
  - ipopt_options.tol: 1×10^{-8}

1.2.2 Results

- Period Error:
  - 158.05 [s]
  - 0.004087 [%]
- Jacobi Constant/Hamiltonian Deviation:
  - Maximum: 1.8987×10^{-10}
  - From C^*: 1.3855×10^{-8}
- Hamiltonian Deviation:
  - Maximum: 5.3502×10^{-10}
- Solution Run Time:
  - 14.4 [s]

1.2.3 Plots

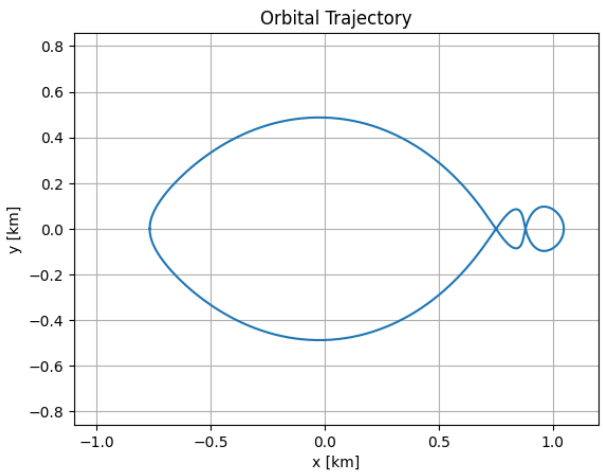

Figure 6: 2D Representation of Orbit (xy-plane)

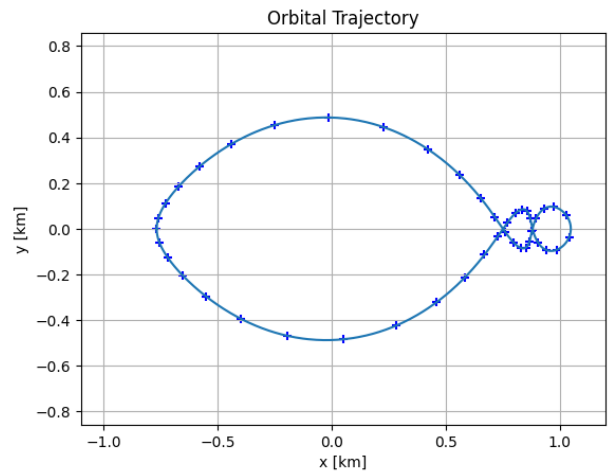

Figure 7: 2D Representation of Orbit with Day Ticks(xy-plane)

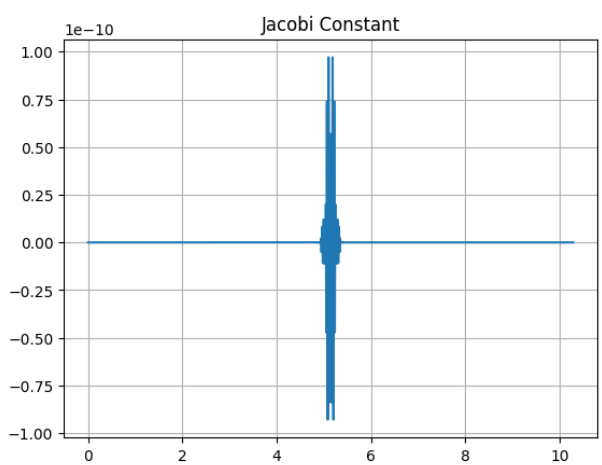

Figure 8: Jacobi Constant over Orbit

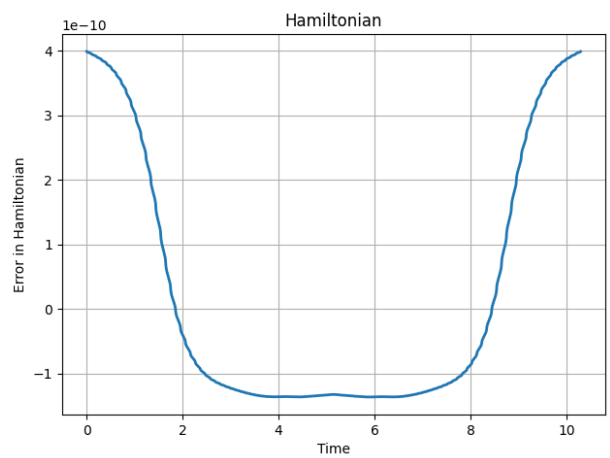

Figure 9: Hamiltonian over Orbit

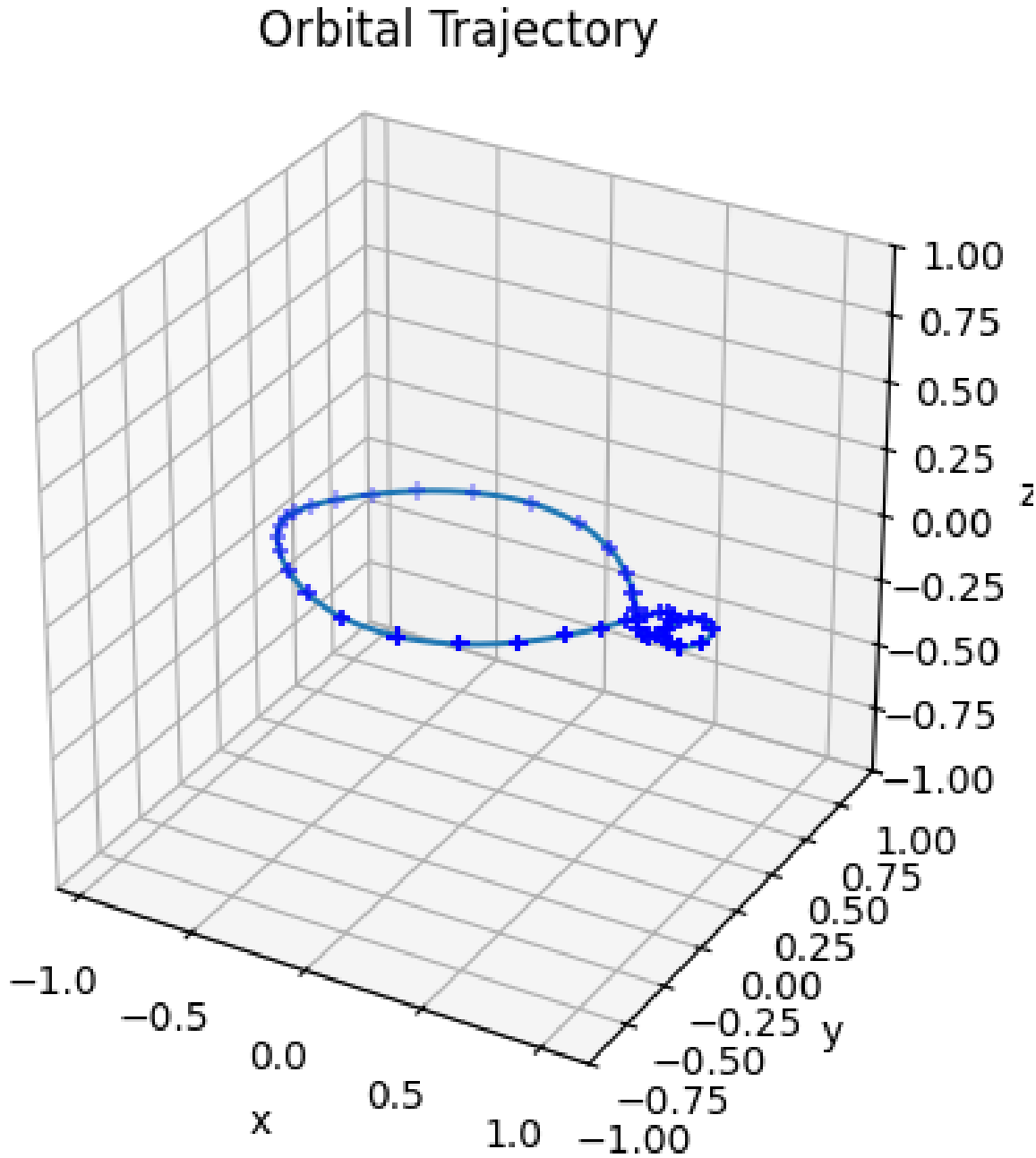

Figure 10: 3D Representation of Orbit

1.3 No Optimization

1.3.1 Setup

- Objective:
  - $J = 0$
- Constraints: Let $\begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix} = \vec{v}_i \times \vec{v}_f$; $\epsilon = 1 \times 10^{-3}$
  - $x_f - x_i = 0$
  - $y_f - y_i = 0$
  - $z_f - z_i = 0$
  - $|c_1| \leq \epsilon$
  - $|c_2| \leq \epsilon$
  - $|c_3| \leq \epsilon$
- Bounds:
  - $t_0 = 0$
- Mesh:
  - Segments: 100
  - Points: 10
- Ipopt Options:
  - ipopt_options.tol: 1×10^{-8}

1.3.2 Results

- Period Error:
  - 158.04 [s]
  - 0.004087 [%]
- Jacobi Constant/Hamiltonian Deviation:
  - Maximum: 1.8987×10^{-10}
  - From C^*: 1.3865×10^{-8}
- Hamiltonian Deviation:
  - Maximum: 1.1635×10^{-12}
- Solution Run Time:
  - 19.0 [s]

1.3.3 Plots

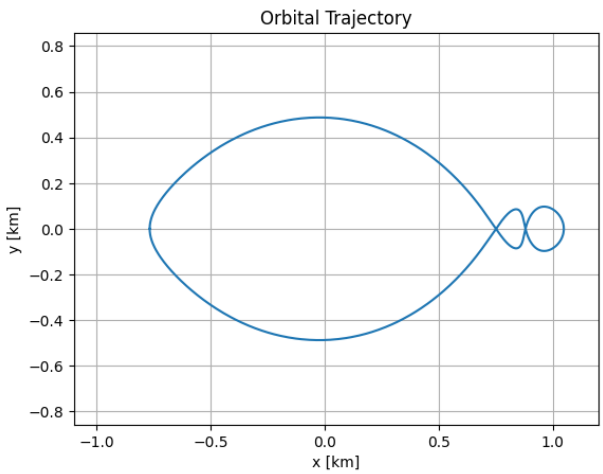

Figure 11: 2D Representation of Orbit (xy-plane)

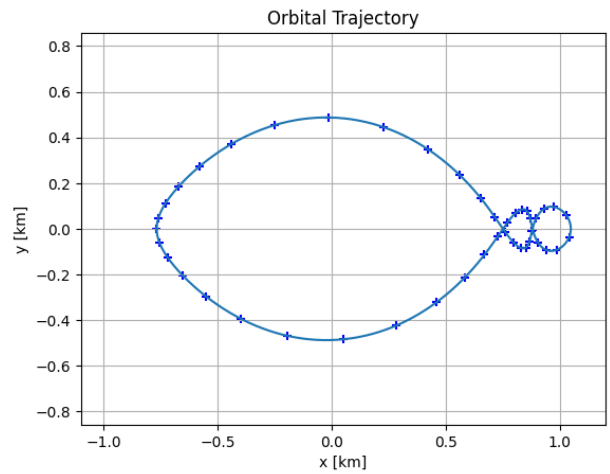

Figure 12: 2D Representation of Orbit with Day Ticks(xy-plane)

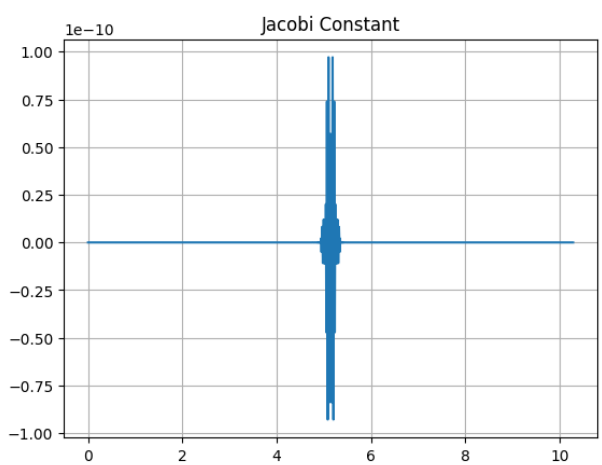

Figure 13: Jacobi Constant over Orbit

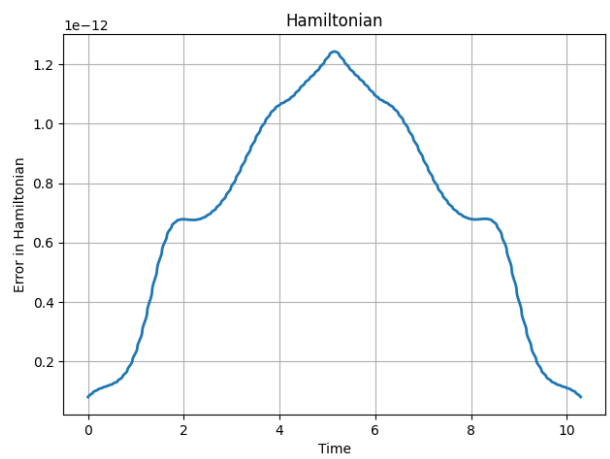

Figure 14: Hamiltonian over Orbit

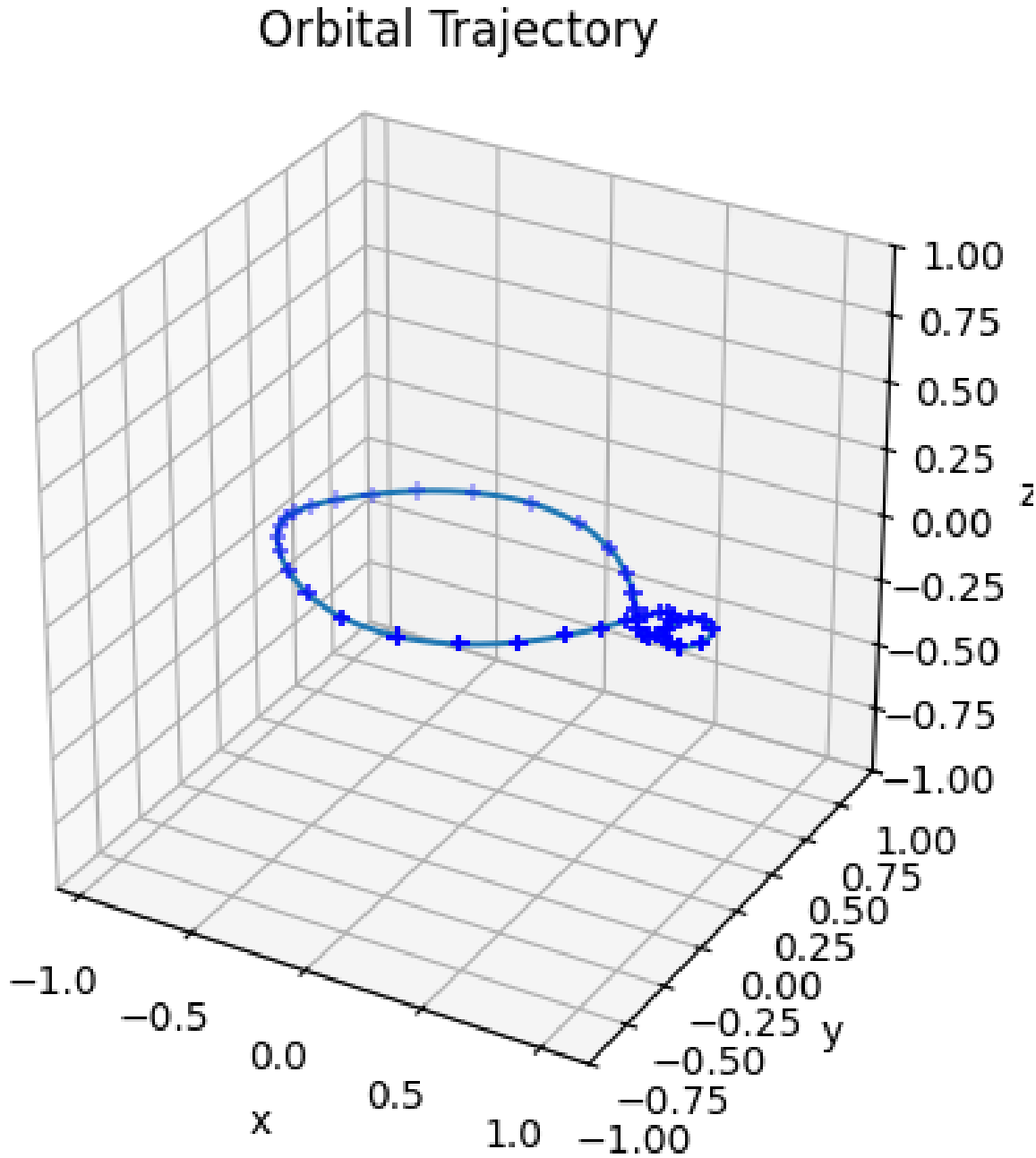

Figure 15: 3D Representation of Orbit

1.4 Minimize Final Time

1.4.1 Setup

- Objective:
  - $J = t_f$
- Constraints: Let $\begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix} = \vec{v}_i \times \vec{v}_f$; $\epsilon = 1 \times 10^{-3}$
  - $x_f - x_i = 0$
  - $y_f - y_i = 0$
  - $z_f - z_i = 0$
  - $|c_1| \leq \epsilon$
  - $|c_2| \leq \epsilon$
  - $|c_3| \leq \epsilon$
- Bounds:
  - $t_0 = 0$
- Mesh:
  - Segments: 100
  - Points: 10
- Ipopt Options:
  - ipopt_options.tol: 1×10^{-8}

1.4.2 Results

- Period Error:
  - 2922571.01 [s]
  - 75.5825 [%]
- Jacobi Constant/Hamiltonian Deviation:
  - Maximum: 80.0983
  - From C^*: 0.5044
- Hamiltonian Deviation:
  - Maximum: 622.3424
- Solution Run Time:
  - 4 [min] 12.5 [s]

1.4.3 Plots

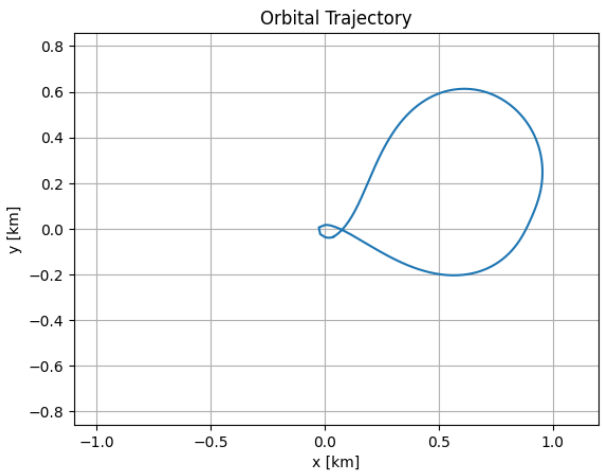

Figure 16: 2D Representation of Orbit (xy-plane)

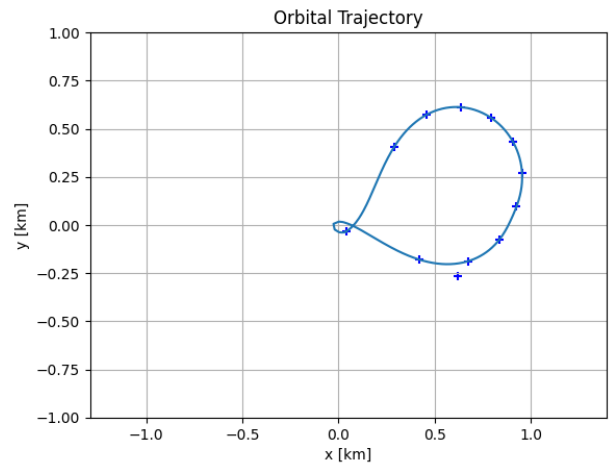

Figure 17: 2D Representation of Orbit with Day Ticks(xy-plane)

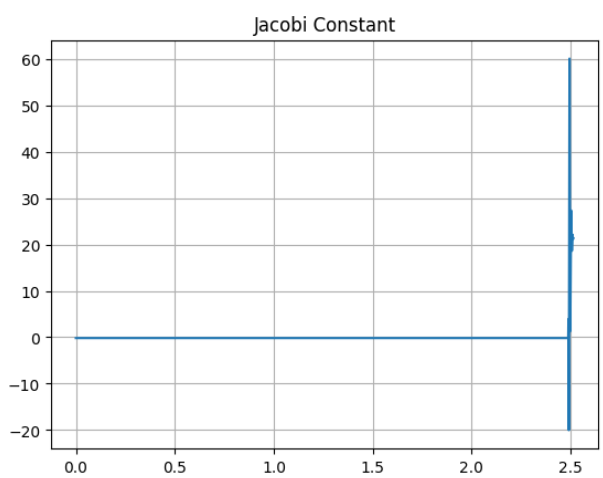

Figure 18: Jacobi Constant over Orbit

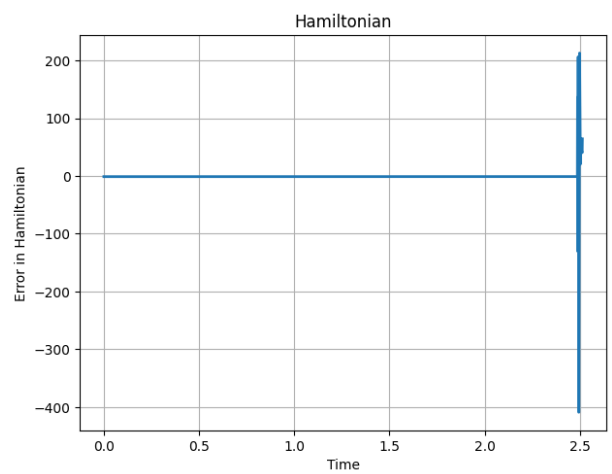

Figure 19: Hamiltonian over Orbit

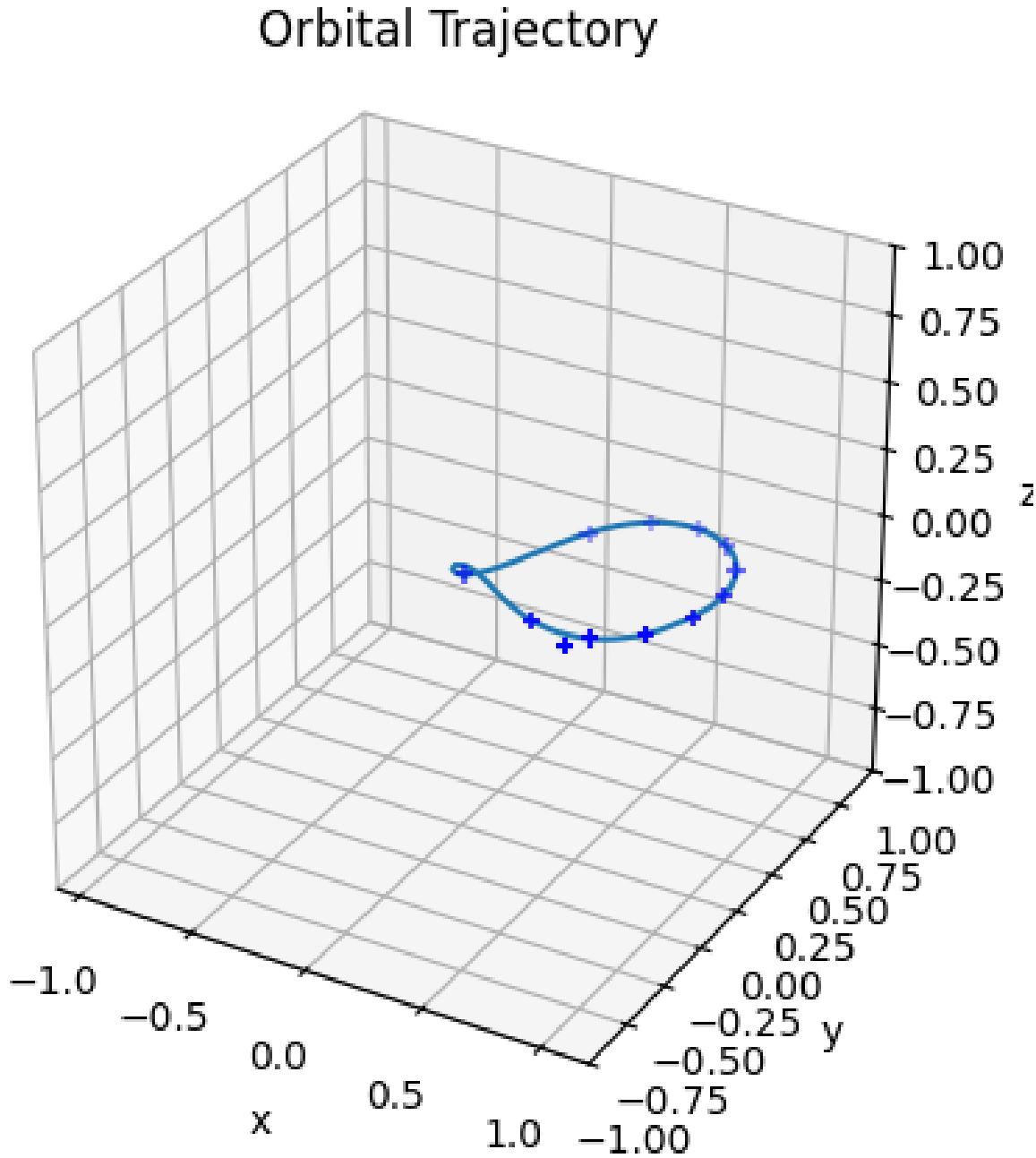

Figure 20: 3D Representation of Orbit

1.5 Cross Product as Objective

1.5.1 Setup

- Objective: Let $\begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix} = \vec{v}_i \times \vec{v}_f$
  - $J = W_{cf} \left(C_f - C^*\right)^2 + W_v \left(c_1^2 + c_2^2 + c_3^2\right)$
- Constraints:
  - $x_f - x_i = 0$
  - $y_f - y_i = 0$
  - $z_f - z_i = 0$
- Bounds:
  - $t_0 = 0$
- Mesh:
  - Segments: 100
  - Points: 10
- Ipopt Options:
  - ipopt_options.tol: 1×10^{-8}

1.5.2 Results

- Period Error:
  - 142.60 [s]
  - 0.003688 [%]
- Jacobi Constant/Hamiltonian Deviation:
  - Maximum: 1.8987×10^{-10}
  - From C^*: 9.9576×10^{-8}
- Hamiltonian Deviation:
  - Maximum: 1.8659×10^{-7}
- Solution Run Time:
  - 8.9 [s]

1.5.3 Plots

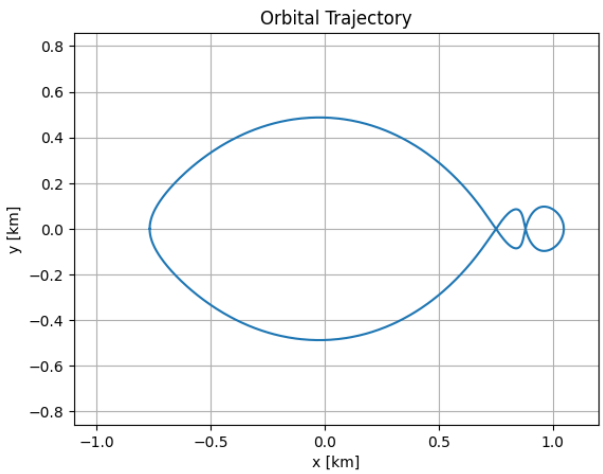

Figure 21: 2D Representation of Orbit (xy-plane)

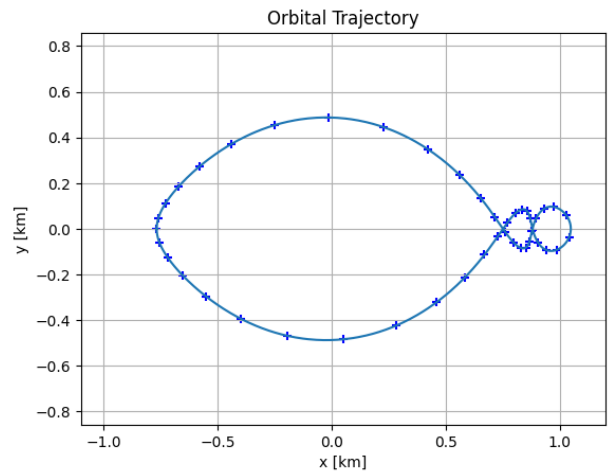

Figure 22: 2D Representation of Orbit with Day Ticks(xy-plane)

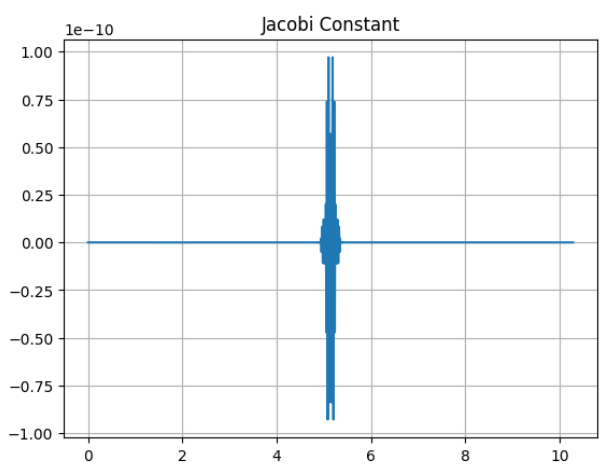

Figure 23: Jacobi Constant over Orbit

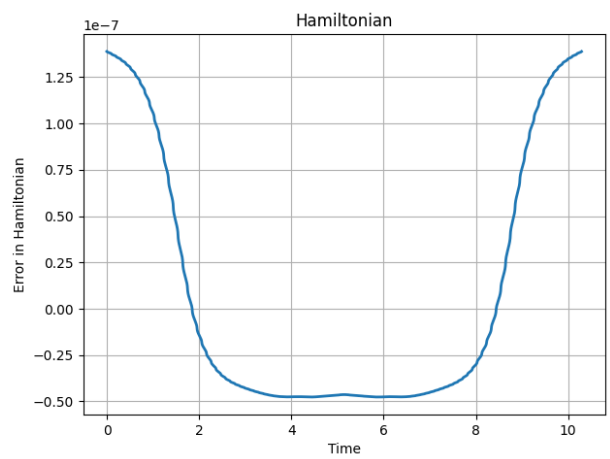

Figure 24: Hamiltonian over Orbit

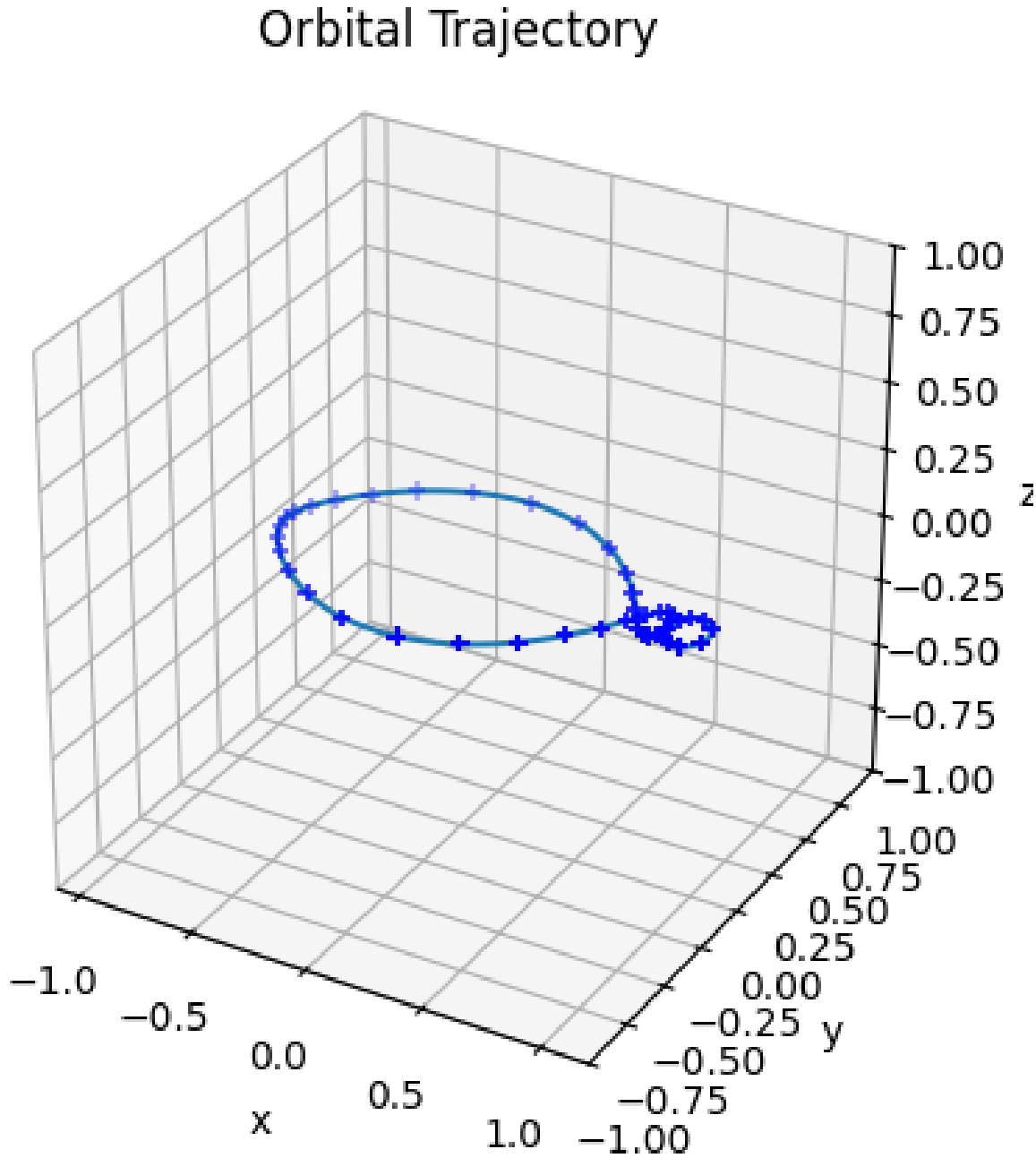

Figure 25: 3D Representation of Orbit

1.6 Cross Product Only

1.6.1 Setup

- Objective: Let $\begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix} = \vec{v}_i \times \vec{v}_f$
  - $J = W_v \left(c_1^2 + c_2^2 + c_3^2\right)$
- Constraints:
  - $x_f - x_i = 0$
  - $y_f - y_i = 0$
  - $z_f - z_i = 0$
- Bounds:
  - $t_0 = 0$
- Mesh:
  - Segments: 100
  - Points: 10
- Ipopt Options:
  - ipopt_options.tol: 1×10^{-8}

1.6.2 Results

- Period Error:
  - 142.60 [s]
  - 0.003688 [%]
- Jacobi Constant/Hamiltonian Deviation:
  - Maximum: 1.8988×10^{-10}
  - From C^*: 9.9635×10^{-8}
- Hamiltonian Deviation:
  - Maximum: 1.8296×10^{-7}
- Solution Run Time:
  - 10.9 [s]

1.6.3 Plots

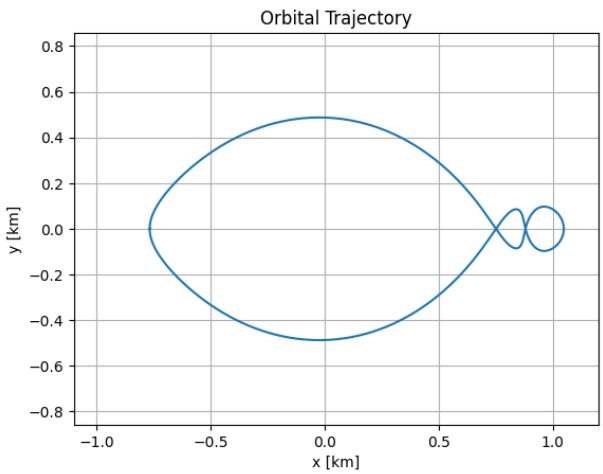

Figure 26: 2D Representation of Orbit (xy-plane)

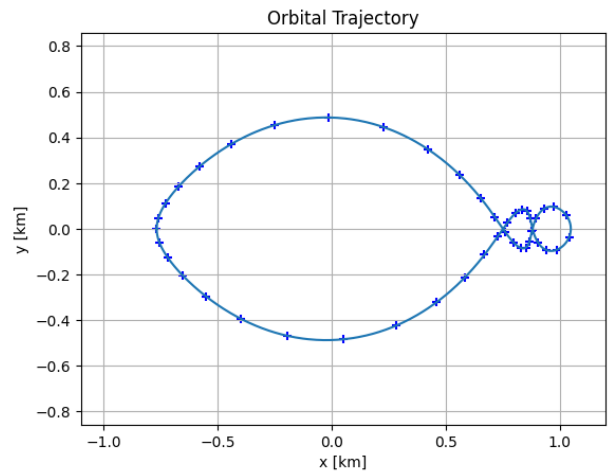

Figure 27: 2D Representation of Orbit with Day Ticks(xy-plane)

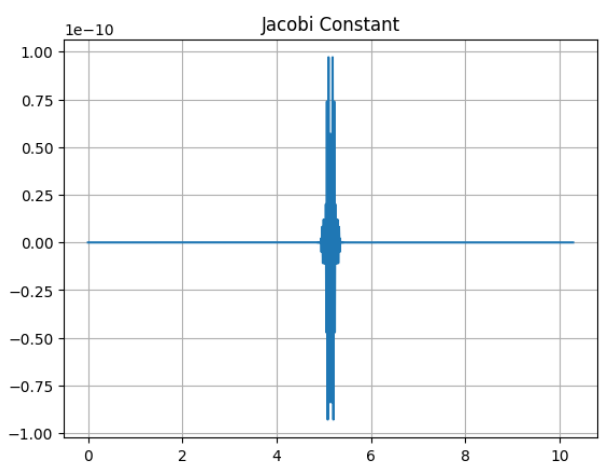

Figure 28: Jacobi Constant over Orbit

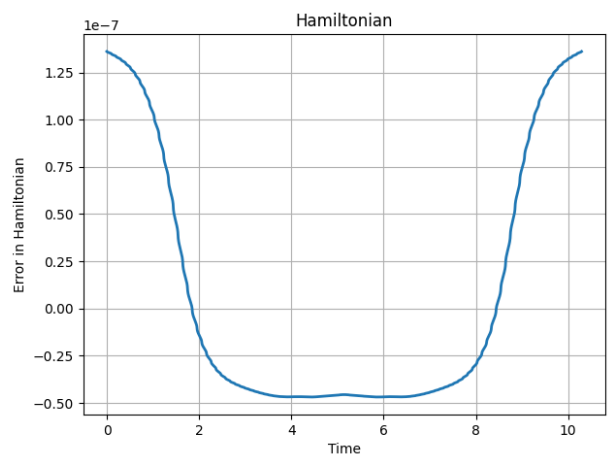

Figure 29: Hamiltonian over Orbit

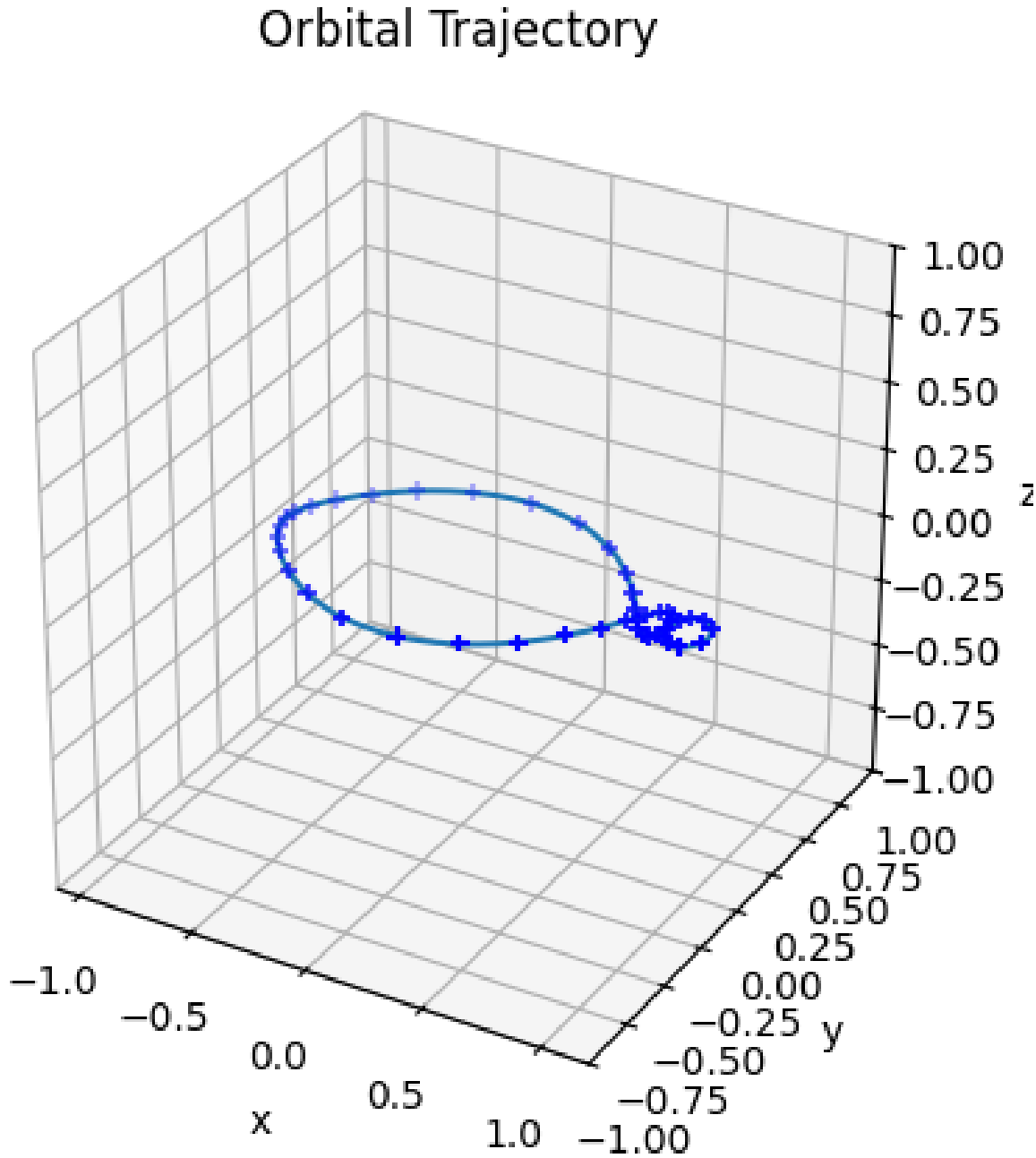

Figure 30: 3D Representation of Orbit

1.7 Notes

- The integral of the difference in Jacobi constant only hurts performance - good idea, but not necessary with cross product constraint
- Objective as just final Jacobi constant or nothing at all both work, keeping the final Jacobi just to have something to minimize seems better
- Minimum time is a trap
- Putting the cross product in the objective instead of in the constraints seems to work well
- Best option for exploring 3D orbits is probably having only the cross product in the objective and not the final Jacobi constant

2 Generating Tilted Orbits for (1,1) Cyclers

2.1 45 Degree Guess

2.1.1 Setup

- Objective: Let $\begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix} = \vec{v}_i \times \vec{v}_f$
  - $J = W_v \left(c_1^2 + c_2^2 + c_3^2\right)$
- Constraints:
  - $x_f - x_i = 0$
  - $y_f - y_i = 0$
  - $z_f - z_i = 0$
- Bounds:
  - $t_0 = 0$
- Mesh:
  - Segments: 100
  - Points: 10
- Ipopt Options:
  - ipopt_options.tol: 1×10^{-8}
- Initial Guess: Take the planar cycler generated from the approach using the cross product as the objective and nothing else and rotate the entire orbit about the x-axis by exactly 45 degrees.

2.1.2 Results

- Jacobi Constant/Hamiltonian Deviation:
  - Maximum: 1.5788×10^{-10}
- Hamiltonian Deviation:
  - Maximum: 9.6679×10^{-9}
- Solution Run Time:
  - 6.8 [s]

2.1.3 Plots

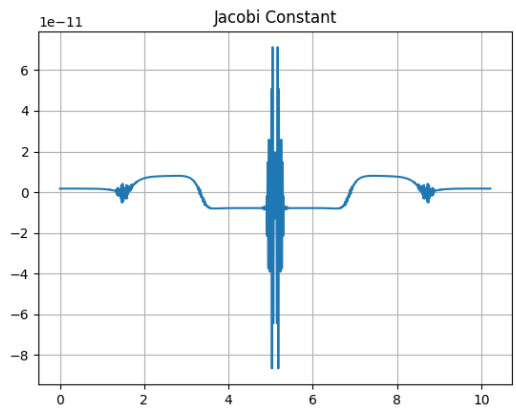

Figure 31: Jacobi Constant over Orbit

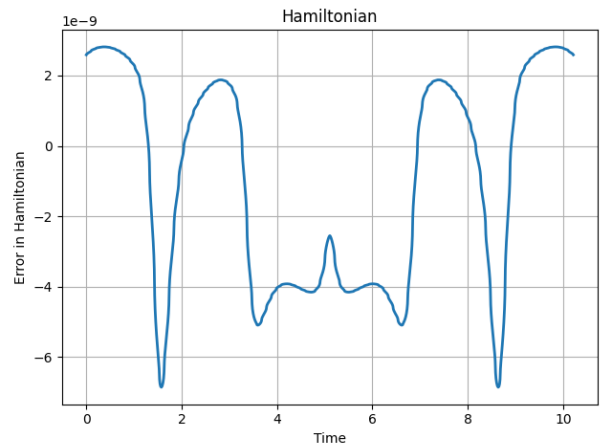

Figure 32: Hamiltonian over Orbit

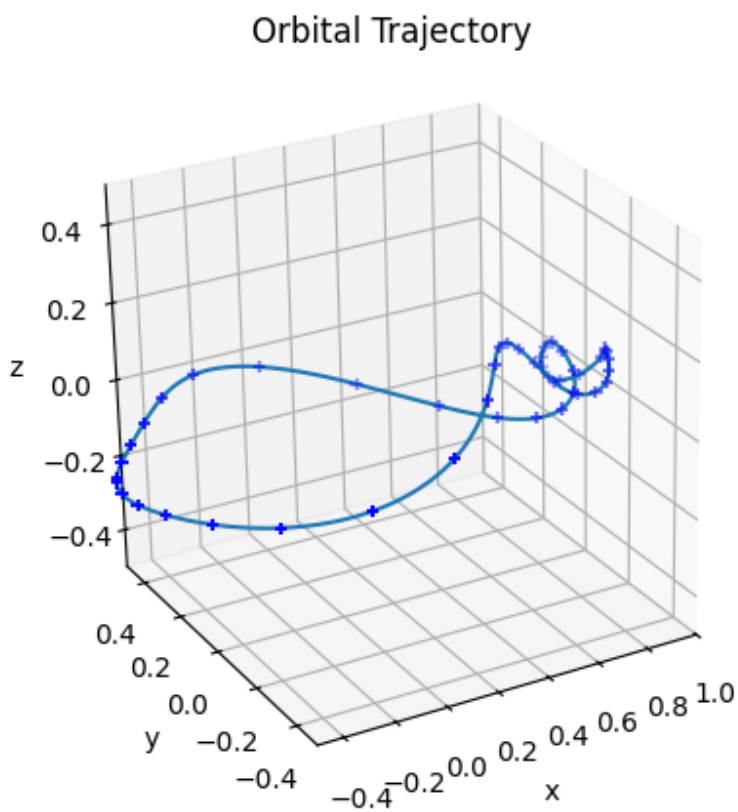

Figure 33: 3D Representation of Orbit

2.2 60 degree Guess

2.2.1 Setup

- Objective: Let $\begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix} = \vec{v}_i \times \vec{v}_f$
  - $J = W_v \left(c_1^2 + c_2^2 + c_3^2\right)$
- Constraints:
  - $x_f - x_i = 0$
  - $y_f - y_i = 0$
  - $z_f - z_i = 0$
- Bounds:
  - $t_0 = 0$
- Mesh:
  - Segments: 100
  - Points: 10
- Ipopt Options:
  - ipopt_options.tol: 1×10^{-8}
- Initial Guess: Take the planar cycler generated from the approach using the cross product as the objective and nothing else and rotate the entire orbit about the x-axis by exactly 45 degrees.

2.2.2 Results

- Jacobi Constant/Hamiltonian Deviation:
  - Maximum: 8.6509×10^{-13}
- Hamiltonian Deviation:
  - Maximum: 2.3155×10^{-10}
- Solution Run Time:
  - 10.5 [s]

2.2.3 Plots

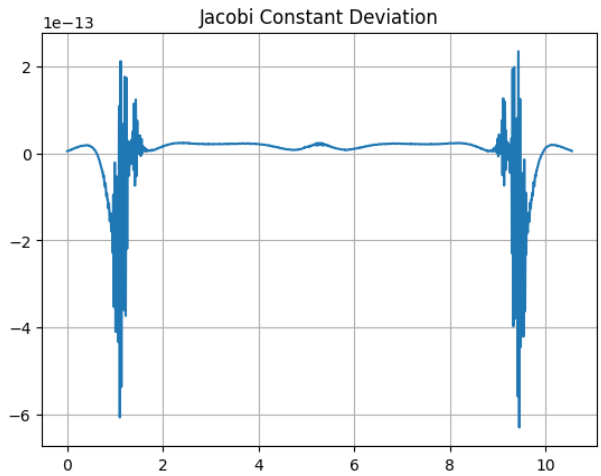

Figure 34: Jacobi Constant over Orbit

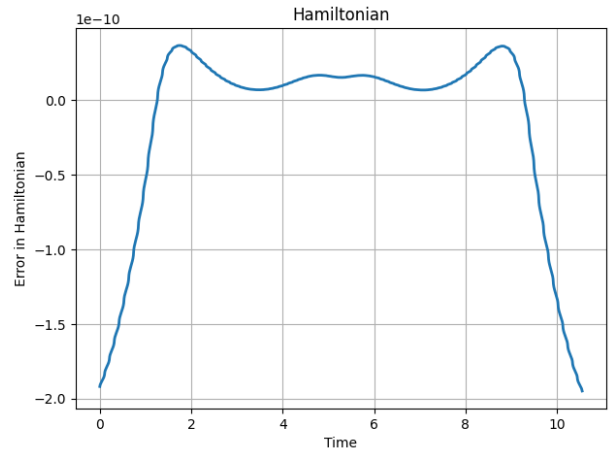

Figure 35: Hamiltonian over Orbit

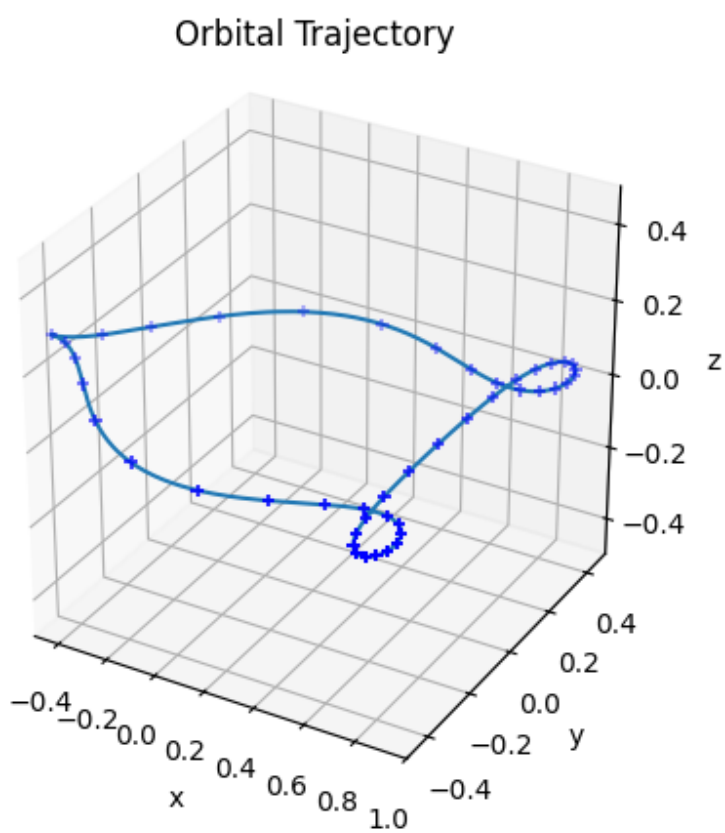

Figure 36: 3D Representation of Orbit

2.3 30 Degree Guess

2.3.1 Setup

- Objective: Let $\begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix} = \vec{v}_i \times \vec{v}_f$
  - $J = W_v \left(c_1^2 + c_2^2 + c_3^2\right)$
- Constraints:
  - $x_f - x_i = 0$
  - $y_f - y_i = 0$
  - $z_f - z_i = 0$
- Bounds:
  - $t_0 = 0$
- Mesh:
  - Segments: 100
  - Points: 10
- Ipopt Options:
  - ipopt_options.tol: 1×10^{-8}
- Initial Guess: Take the planar cycler generated from the approach using the cross product as the objective and nothing else and rotate the entire orbit about the x-axis by exactly 45 degrees.

2.3.2 Results

- Jacobi Constant/Hamiltonian Deviation:
  - Maximum: 1.5788×10^{-10}
- Hamiltonian Deviation:
  - Maximum: 9.6679×10^{-9}
- Solution Run Time:
  - 6.8 [s]

2.3.3 Plots

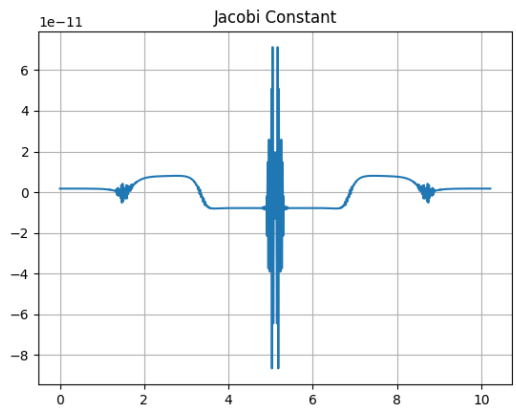

Figure 37: Jacobi Constant over Orbit

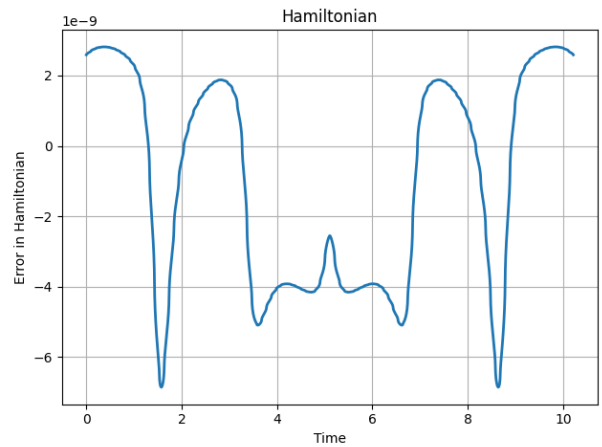

Figure 38: Hamiltonian over Orbit

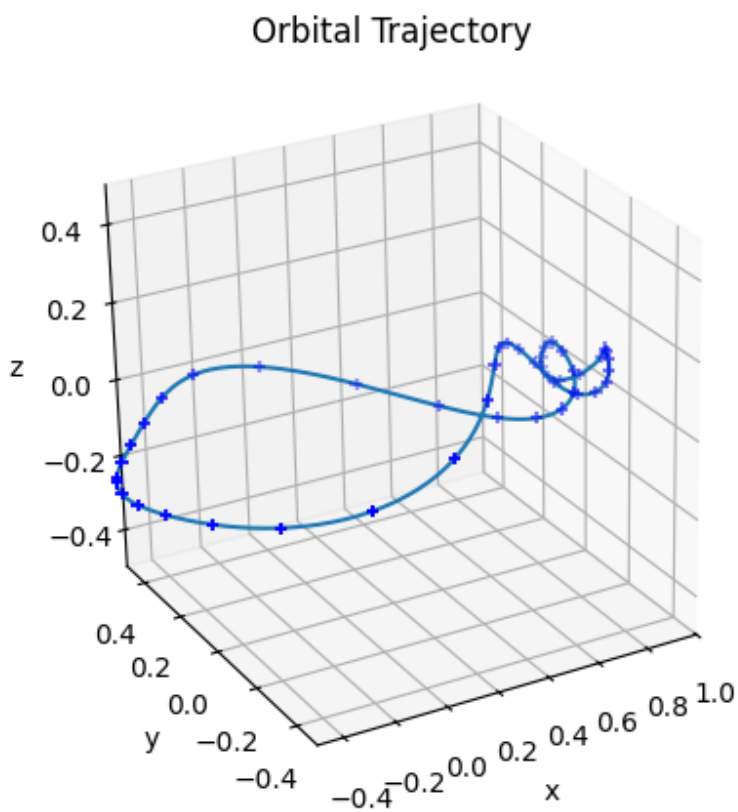

Figure 39: 3D Representation of Orbit

2.4 Notes

The 45 degree guess produced a feasible inclined orbit, and the 60 degree guess produced an orbit which is not inclined, but is also feasible and potentially indicates a family of pseudocyclers. Next steps are to perform an angle sweep to find out which angles give good orbits, recycling the solutions as guesses up to 90 degrees of inclination. The problem setup is much better, yielding better results with significantly decreased run times. After the angle sweep, plan is to look into other cycler families and solar perturbations.